

Himanshu Singh

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PROFESSIONAL SUMMARY

AI SME / Team Lead – AI Engineer with **5 years of experience** delivering **AI and GenAI solutions** for **U.S. healthcare and Fortune 100 clients**. Experienced in **AI solution architecture, client engagement, AI agents, RAG, OCR, document intelligence, LLM applications**, and workflow automation using **Azure AI Foundry, LangChain, LangGraph, Python, Django, FastAPI, and Azure services**. Proven expertise in leading teams, translating business requirements into scalable AI solutions, and leveraging a strong **data engineering background** across Azure, Databricks, Spark, Hadoop, and large-scale data platforms.

EXPERIENCE

TATA CONSULTANCY SERVICES

TCS Healthcare Innovation Lab (Nov 2025 – Present)

AI SME – TL

Hyderabad, India

- Leading a team within the **TCS Healthcare Innovation Lab**, partnering with **U.S. healthcare clients** to understand business requirements, design solution architectures, and deliver high-volume **AI solutions** across diverse healthcare use cases.
- Engaging directly with client stakeholders to **gather requirements, translate business problems into technical solutions, and lead end-to-end delivery** in collaboration with the engineering team.
- Architecting and delivering diverse AI applications including **OCR, medical document extraction, intelligent text processing, information extraction, and workflow automation** using **Azure AI Foundry, LangChain**, and Azure AI services.
- Developing scalable full-stack AI applications using **Azure AI Foundry, LangChain, Azure Functions, Django, React**, and Azure cloud services, integrating AI capabilities into enterprise healthcare workflows.
- Delivered a portfolio of **AI-powered healthcare solutions** including **Auto RFP, MR Extractor, handwritten prescription OCR, and discharge bill processing**, along with numerous client-specific AI use cases.
- Driving AI solution development from **client discovery and architecture design through prototyping, development, and delivery**, while guiding the team on AI technologies, solution design, and implementation.

TATA CONSULTANCY SERVICES

KAISER PERMANENTE (Mar 2023 – Present)

Data Engineer

Hyderabad, India

- Delivered **20TB+ of healthcare data** into **Azure Data Lake Storage (ADLS)** by **designing and optimizing enterprise-scale data pipelines** using **Azure Data Factory (ADF)** and **Databricks Delta Lake**, enabling faster analytics and reporting.
- Increased data processing efficiency by **60%** by **automating ETL/ELT workflows** across **ADF, ADLS, Databricks, and PySpark**, ensuring near **real-time data availability**.
- Improved data accuracy by **100%** through **implementing robust validation and quality frameworks**, delivering highly reliable and actionable business insights.
- Enabled seamless enterprise-wide analytics by **integrating on-premise, third-party, and Hadoop data sources (Oracle, Taleo, JSON, Avro, CSV, Parquet)** into ADLS via **ADF connectors**.
- Reduced Spark job runtime by **20%** for **2M+ daily records** by **implementing Spark performance tuning techniques** (caching, partitioning, broadcast joins) across **Hadoop environments**.
- Optimized **on-premise Hadoop clusters (HDFS, Hive, Spark)** by **fine-tuning resource allocation and data partitioning strategies**, improving large-scale batch processing performance by **35%** and ensuring smooth integration with **Azure cloud migration pipelines**.

TATA CONSULTANCY SERVICES

WALGREENS (Oct 2021 – Feb 2023)

Data Engineer

Noida, India

- Accelerated large-scale text processing by **90%** (2 hours → 15 minutes) by **developing and optimizing MapReduce programs** in the **Hadoop ecosystem**, significantly improving ETL throughput.
- Enhanced operational efficiency by **50%** by **automating data workflows** using **Python, HiveQL, and Shell scripting**, reducing manual intervention and improving reliability.

- Improved analytical data trustworthiness by **30%** through **data quality, consistency, and reconciliation checks** using **Hive, Impala, and Spark**.
- Streamlined ETL orchestration and reduced manual efforts by **40%** by **automating complex data pipelines** with **Python, HiveQL, and Shell**, achieving a **60% improvement in throughput**.
- Supported cloud migration initiatives by **managing data ingestion and batch processing in Hadoop Distributed File System (HDFS)**, laying the foundation for **Azure-based architectures**.
- Delivered enterprise-wide reporting solutions by **collaborating with business teams** and **processing large-scale datasets** across distributed ETL systems.

OPEN SOURCE CONTRIBUTIONS

Liu Embeddings	[PyPI ↗] [Docs ↗]
Developed and published an open-source Python library, liuembeddings , which combines text embeddings and vector storage for efficient local NLP retrieval tasks. Built on Hugging Face Transformers and ChromaDB , it offers a free, fully local, and privacy-friendly alternative to cloud-based embedding services ideal for AI projects requiring fast and cost-efficient embedding, storage, and retrieval workflows.	
DbMeta	[PyPI ↗] [Git ↗]
Developed and published an open-source, zero-dependency Python engine, DbMeta , designed for parsing and analyzing batch job metadata from Hadoop, Spark, Databricks by using folder-based job-run structures. Automatically detects metadata files, extracts structured metadata, and converts them into queryable tables for efficient monitoring and analytics.	

TECHNICAL SKILLS

AI & GenAI:	Azure AI Foundry, LangChain, LangGraph, Prompt Engineering, Open-Source Models
Backend & APIs:	Python, Django, Django REST Framework, FastAPI
Cloud & Azure:	Azure Functions, Azure Blob Storage, Azure Queue Storage, ADLS, Azure DevOps
Vector & DataStores:	MongoDB, SQL, PostgreSQL, MySQL, Oracle
Data Architecture:	ETL/ELT, Data Modelling, Data Warehousing, Data Lakehouse, OLAP/OLTP
Data Engineering:	Databricks, Apache Spark (PySpark, RDD, Streaming), Hadoop, Hive, Impala, MapReduce, ADF, Azure Synapse

CERTIFICATIONS

Databricks Fundamentals 2025	View Certificate ↗
Databricks Certified Data Engineer Associate 2025	View Certificate ↗
Azure Fundamentals (TCS Digital) 2025	View Certificate ↗
Generative AI (Udemy) 2025	View Certificate ↗
Python Programming (TCS Digital) 2022	View Certificate ↗

PROJECTS

LeetCode SQL to PySpark (Top 50 Questions)	View on GitHub ↗
DataLemur SQL to PySpark	View on GitHub ↗
DataLemur SQL to PySpark (RDD Implementation)	View on GitHub ↗
Databricks Ecommerce analytics	View on GitHub ↗
Databricks ETL	View on GitHub ↗
AI Chatbot	View on GitHub ↗
The Analysis Bot	View on GitHub ↗
AI DATA Engineer Copilot	View on GitHub ↗

EDUCATION

NIET	Greater Noida, UP
<i>Bachelor of Technology in Computer Science & Engineering</i>	<i>Aug 2017 – Sep 2021</i>

ABC Public School
Senior Secondary School

Gorakhpur, UP
Apr 2015 – Mar 2017